

elements. Also describe any five built-in dictionary methods.

4. Explain the concept of iterators and generators in python. Discuss how iter() and next() methods work in iterator. Also explain generator functions and the yield statement with example.
5. Write a program to enter N strings from user in a list and count how many strings are of length more than or equal to 2 and their first and last letters are same.
6. Write a program to enter a string and check the string is palindrome or not using user define function.
7. What is function in python ? Explain function definition, calling and return statement. Also describe different types of function arguments with suitable examples.
8. What is file handling in python ? Explain text vs binary files and various file modes. Also describe the use of read(), readline(), write(), close(), seek() and tell() with suitable explanations.

2025

Time : 3 hours

Full Marks : 70

Candidates are required to give their answers in their own words as far as practicable.

The figures in the margin indicate full marks.

*Answer from **all** the Sections as directed.*

(Python Programming)

Section – A

1. [A] Choose the correct answer of the following :

1×5 = 5

(a) What will list 1[::-1] do ?

(i) Sort the list

(ii) Reverse the list

(iii) Duplicate the list

(iv) Remove last element

(b) Which of the following set operations returns new set without modifying originals ?

- ~~(i) union()~~ (ii) add()
(iii) discard() (iv) clear()

(c) Which of the following is a tuple ?

- (i) [1, 2, 3] (ii) {1, 2, 3}
~~(iii) (1, 2, 3)~~ (iv) {}

(d) What will be the value of the following python expression $4 + 3\%2$?

- (i) 7 (ii) 2
~~(iii) 5~~ (iv) 1

(e) Which of the following is the truncation division operator in python ?

- ~~(i) //~~ (ii) /
(iii) | (iv) %

[B] Fill in the blanks : $1 \times 5 = 5$

(a) A _____ is an ordered, immutable collection of elements.

(b) The method _____ is used to convert all letters of a string into lowercase.

(c) Python is a _____ language.

(d) The method _____ is used to remove a specific key-value pair from a dictionary.

(e) The function _____ is used to sort elements of a list in ascending order.

Section – B

2. Answer any **four** questions of the following :

$5 \times 4 = 20$

(a) What is the different types of string operations available in python such as concatenation, slicing and repetition.

(b) Write a program to input an integer N and print it is even or odd.

(c) Write a program to print largest number among three given number using if-elif-else.

(d) Discuss built in datatypes in python.

(e) What are python modules ? Give a program to demonstrate the used defined module.

(f) What are local variable and global variable in python ? Explain with program.

Section – C

Answer any **four** questions of the following :

$10 \times 4 = 40$

8. What is dictionary in python ? Explain how to create, access update and delete dictionary